Exercise 10.3 SOLUTION

Q1. Find the values of $\sin 2\alpha$, $\cos 2\alpha$ and $\tan 2\alpha$, when:

i. $\sin\alpha = \frac{3}{5};\ 0 < \alpha < \frac{\pi}{2}$

Sol: $\cos^2\alpha = 1 - \sin^2\alpha$

$= 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25} = \frac{16}{25}$

$\cos\alpha = \pm\frac{4}{5} \Rightarrow \cos\alpha = \frac{4}{5}\ (\alpha\ is\ in\ I\ quad)$

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = \frac{3/5}{4/5} = \frac{3}{4}$

Now

(a) $\sin 2\alpha = 2\sin\alpha\cos\alpha$

$= 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$

(b) $\cos 2\alpha = 2\cos^2\alpha - 1 = 2\left(\frac{4}{5}\right)^2 - 1 = 2\left(\frac{16}{25}\right) - 1$

$= \frac{32-25}{25} = \frac{7}{25}$

(c) $\tan 2\alpha = \frac{\sin 2\alpha}{\cos 2\alpha} = \frac{24/25}{7/25} = \frac{24}{25} \times \frac{25}{7} = \frac{24}{7}$

ii. $\cos\alpha = \frac{4}{5};\ 0 < \alpha < \frac{\pi}{2}$

Sol: $\sin^2\alpha = 1 - \cos^2\alpha$

$= 1 - \left(\frac{4}{5}\right)^2 = 1 - \frac{16}{25} = \frac{9}{25}$

$\sin\alpha = \pm\frac{3}{5} \Rightarrow \sin\alpha = \frac{3}{5}(\alpha\ is\ in\ I)$

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = \frac{3/5}{4/5} = 3/4$

(a) $\sin 2\alpha = 2\sin\alpha\cos\alpha$

$= 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$

(b) $\cos 2\alpha = 2\cos^2\alpha - 1$

$= 2\left(\frac{4}{5}\right)^2 - 1 = 2\left(\frac{16}{25}\right) - 1$

$= \frac{32-25}{25} = \frac{7}{25}$

(c) $\tan 2\alpha = \frac{\sin 2\alpha}{\cos 2\alpha} = \frac{24/25}{7/25} = \frac{24}{7}$

2. Prove the following identities

i. $\mathbf{Cot\,\alpha - \tan\alpha = 2Cot2\,\alpha}$

Sol: L.H.S = $Cot\,\alpha - \tan\alpha$

$= \frac{Cos\alpha}{Sin\alpha} - \frac{Sin\alpha}{Cos\alpha} = \frac{Cos^2\alpha - Sin^2\alpha}{Sin\alpha Cos\alpha}$

$= \frac{2(Cos^2\alpha - Sin^2\alpha)}{2Sin\alpha Cos\alpha}$ ('X' & ÷ by 2)

$= \frac{2Cos2\alpha}{Sin2\alpha} = 2\ Cot2\,\alpha$ = R.H.S

ii. $\frac{Sin2\alpha}{1+Cos2\alpha} = \tan\alpha$

Sol. L.H.S = $\frac{Sin2\alpha}{1+Cos2\alpha}$

$= \frac{2Sin\alpha Cos\alpha}{2Cos^2\alpha} = \frac{Sin\alpha}{Cos\alpha} = \tan\alpha$

= R.H.S

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iii. $\frac{1-Cos\alpha}{Sin\alpha} = \tan\frac{\alpha}{2}$

Sol. $L.H.S = \frac{1-Cos\alpha}{Sin\alpha} = \frac{2Sin^2\frac{\alpha}{2}}{2Sin\frac{\alpha}{2}Cos\frac{\alpha}{2}} = \frac{Sin\frac{\alpha}{2}}{Cos\frac{\alpha}{2}}$

$= \tan\frac{\alpha}{2} = R.H.S$

iv. $\frac{Cos\alpha - Sin\alpha}{Cos\alpha + Sin\alpha} = Sec2\alpha - \tan2\alpha$

Sol. $L.H.S = \frac{Cos\alpha - Sin\alpha}{Cos\alpha + Sin\alpha}$

$= \frac{Cos\alpha - Sin\alpha}{Cos\alpha + Sin\alpha} \times \frac{Cos\alpha - Sin\alpha}{Cos\alpha - Sin\alpha}$

$= \frac{(Cos\alpha - Sin\alpha)^2}{Cos^2\alpha - Sin^2\alpha} =$

$\frac{Cos^2\alpha + Sin^2\alpha - 2SinaCos\alpha}{Cos2\alpha} = \frac{1-Sin2\alpha}{Cos2\alpha}$

$= \frac{1}{Cos2\alpha} - \frac{Sin2\alpha}{Cos2\alpha} = Sec2\alpha - \tan2\alpha = R.H.S$

$= Sec2\alpha - \tan2\alpha = R.H.S$

v. $\sqrt{\frac{1+Sin\alpha}{1-Sin\alpha}} = \frac{Sin\alpha/2 + Cos\alpha/2}{Sin\alpha/2 - Cos\alpha/2}$

Sol. $L.H.S = \sqrt{\frac{1+Sin\alpha}{1-Sin\alpha}}$

$= \sqrt{\frac{Sin^2\alpha/2 + Cos^2\alpha/2 + 2Sin\alpha/2Cos\alpha/2}{Sin^2\alpha/2 + Cos^2\alpha/2 - 2Sin\alpha/2Cos\alpha/2}}$

$= \sqrt{\frac{(Sin\alpha/2 + Cos\alpha/2)^2}{(Sin\alpha/2 - Cos\alpha/2)^2}} = \frac{Sin\alpha/2 + Cos\alpha/2}{Sin\alpha/2 - Cos\alpha/2}$

$= R.H.S$

vi. $\frac{Cosec\theta + 2Cosec2\theta}{Sec\theta} = Cot\theta/2$

Sol. $L.H.S = \frac{Cosec\theta + 2Cosec2\theta}{Sec\theta} = \frac{\frac{1}{Sin\theta} + \frac{2}{Sin2\theta}}{\frac{1}{Cos\theta}}$

$= \frac{\frac{1}{Sin\theta} + \frac{2}{2Sin\theta Cos\theta}}{\frac{1}{Cos\theta}} = \left(\frac{Cos\theta + 1}{Sin\theta Cos\theta}\right)\frac{Cos\theta}{1}$

$= \frac{2Cos^2\theta/2}{2Sin\theta/2Cos\theta/2} = \frac{Cos\theta/2}{Sin\theta/2}$

$= Cot\theta/2 = R.H.S$

vii. $1 + \tan\alpha \tan2\alpha = Sec2\alpha$

Sol. $L.H.S = 1 + \tan\alpha \tan2\alpha$

$= 1 + \frac{Sin\alpha}{Cos\alpha} \cdot \frac{Sin2\alpha}{Cos2\alpha} = \frac{CosaCos2\alpha + SinaSin2\alpha}{CosaCos2\alpha}$

$= \frac{CosaCos2\alpha + SinaSin2\alpha}{CosaCos2\alpha}$

$= \frac{Cos(2\alpha - \alpha)}{Cos\alpha \times Cos2\alpha} = \frac{Cos\alpha}{Cos\alpha \times Cos2\alpha}$

$= \frac{1}{Cos2\alpha} = Sec2\alpha = R.H.S$

viii. $\frac{2Sin\theta\, Sin2\theta}{Cos\theta + Cos3\theta} = \tan2\theta \tan\theta$

Sol. $L.H.S = \frac{2Sin\theta\, Sin2\theta}{Cos\theta + Cos3\theta}$ $(Cos3\theta = 4Cos^3\theta - 3Cos\theta)$

$= \frac{2Sin\theta\, Sin2\theta}{Cos\theta + 4Cos^3\theta - 3Cos\theta}$

$= \frac{2Sin\theta\, Sin2\theta}{4Cos^3\theta - 2Cos\theta} = \frac{2Sin\theta\, Sin2\theta}{2Cos\theta(2Cos^2\theta - 1)}$

$= \frac{Sin\theta . Sin2\theta}{Cos\theta . Cos2\theta} = \tan\theta\tan2\theta$

$= \tan2\theta \,.\, \tan\theta = R.H.S$

ix. $\frac{Sin3\theta}{Sin\theta} - \frac{Cos3\theta}{Cos\theta} = 2$

Sol. $L.H.S = \frac{Sin3\theta}{Sin\theta} - \frac{Cos3\theta}{Cos\theta}$

$= \frac{Sin3\theta Cos\theta - Cos3\theta Sin\theta}{Sin\theta Cos\theta} = \frac{Sin(3\theta - \theta)}{Sin\theta Cos\theta}$

$= \frac{Sin2\theta}{Sin\theta Cos\theta} = \frac{2\,Sin\theta Cos\theta}{Sin\theta Cos\theta} = 2 = R.H.S$

x. $\frac{Cos3\theta}{Cos\theta} + \frac{Sin3\theta}{Sin\theta} = 4\,Cos2\theta$

Sol. $L.H.S = \frac{Cos3\theta}{Cos\theta} + \frac{Sin3\theta}{Sin\theta} = \frac{Sin\theta Cos3\theta + Cos\theta Sin3\theta}{Cos\theta Sin\theta}$

$= \frac{Sin(\theta + 3\theta)}{Sin\theta Cos\theta} = \frac{2Sin4\theta}{2Sin\theta Cos\theta} = \frac{2Sin2(2\theta)}{Sin2\theta}$

$= \frac{2.2\,Sin2\theta Cos2\theta}{Sin2\theta} = 4\,Cos2\theta$

$= R.H.S$

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xi. $\dfrac{\tan\theta/2 + Cot\theta/2}{Cot\theta/2 - \tan\theta/2} = \text{Sec}\,\theta$

Sol. $\text{L.H.S} = \dfrac{\tan\theta/2 + Cot\theta/2}{Cot\theta/2 - \tan\theta/2}$

$$= \frac{\dfrac{Sin\theta/2}{Cos\theta/2} + \dfrac{Cos\theta/2}{Sin\theta/2}}{\dfrac{Cos\theta/2}{Sin\theta/2} - \dfrac{Sin\theta/2}{Cos\theta/2}} = \frac{\dfrac{Sin^2\theta/2 + Cos^2\theta/2}{Sin\theta/2\,Cos\theta/2}}{\dfrac{Cos^2\theta/2 - Sin^2\theta/2}{Sin\theta/2\,Cos\theta/2}}$$

$$= \frac{1}{Cos\theta/2\,Sin\theta/2} \times \frac{Sin\theta/2\,Cos\theta/2}{Cos^2\theta/2 - Sin^2\theta/2}$$

$$= \frac{1}{Cos\theta} = \text{Sec}\,\theta = \text{R.H.S}$$

xii. $\dfrac{Sin3\theta}{Cos\theta} + \dfrac{Cos3\theta}{Sin\theta} = 2\text{Cot}2\theta$

Sol. $\text{L.H.S} = \dfrac{Sin3\theta}{Cos\theta} + \dfrac{Cos3\theta}{Sin\theta}$

$$= \frac{Sin3\theta\,Sin\theta + Cos3\theta\,Cos\theta}{Sin\theta\,Cos\theta}$$

$$= \frac{Cos3\theta\,Cos\theta + Sin3\theta\,Sin\theta}{Sin\theta\,Cos\theta}$$

$$= \frac{2Cos(3\theta - \theta)}{2Sin\theta\,Cos\theta} = \frac{2Cos2\theta}{Sin2\theta} = 2\,\text{Cot}\,2\theta$$

xiii. $\dfrac{3+\cos4\theta}{1-\cos4\theta} = \dfrac{1}{2}\left(\tan^2\theta + \cot^2\theta\right)$

Sol: $\text{R.H.S} = \dfrac{1}{2}\left(\tan^2\theta + \cot^2\theta\right)$

$$= \frac{1}{2}\left(\frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\sin^2\theta}\right) = \frac{1}{2}\left(\frac{\sin^4\theta + \cos^4\theta}{\sin^2\theta\cos^2\theta}\right) = \frac{1}{2}\left(\frac{\left(\sin^2\theta\right)^2 + \left(\cos^2\theta\right)^2}{\sin^2\theta\cos^2\theta}\right) \quad \because \quad \sin^2\theta = \frac{1-\cos2\theta}{2}$$

$$= \frac{1}{2}\left(\frac{\left(\dfrac{1-\cos2\theta}{2}\right)^2 + \left(\dfrac{1+\cos2\theta}{2}\right)^2}{\sin^2\theta\cos^2\theta}\right)$$

$$\boxed{\cos^2\theta = \frac{1+\cos2\theta}{2}}$$

$$= \frac{1}{2\sin^2\theta\cos^2\theta}\left(\frac{1+\cos^2 2\theta - 2\cos2\theta}{4} + \frac{1+\cos^2 2\theta + 2\cos2\theta}{4}\right)$$

$$= \frac{1}{2\sin^2\theta\cos^2\theta}\left(\frac{1+\cos^2 2\theta \cancel{-2\cos2\theta} + 1 + \cos^2 2\theta + \cancel{2\cos2\theta}}{4}\right)$$

$$= \frac{1}{2\sin^2\theta\cos^2\theta}\left(\frac{2+2\cos^2 2\theta}{4}\right) = \frac{1}{2\sin^2\theta\cos^2\theta}\left(\frac{\cancel{2}\left(1+\cos^2 2\theta\right)}{\cancel{4}_2}\right)$$

$$= \frac{1}{4\sin^2\theta\cos^2\theta}\left(1 + \left(\frac{1+\cos4\theta}{2}\right)\right)$$

$$= \frac{1}{4\sin^2\theta\cos^2\theta}\left(\frac{2+1+\cos4\theta}{2}\right) = \frac{1}{8\left(\dfrac{1-\cos2\theta}{2}\right)\left(\dfrac{1+\cos2\theta}{2}\right)}\left(3+\cos4\theta\right)$$

$$= \frac{3+\cos4\theta}{2\left(1-\cos^2 2\theta\right)} = \frac{3+\cos4\theta}{2\left(1 - \left(\dfrac{1+\cos4\theta}{2}\right)\right)} = \frac{3+\cos4\theta}{\cancel{2}\left(\dfrac{2-1-\cos4\theta}{\cancel{2}}\right)} = \frac{3+\cos4\theta}{1-\cos4\theta} = \text{L.H.S}$$

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xiv. $\dfrac{1+\sin 2\theta}{1-\sin 2\theta} = \tan^2\left(\dfrac{\pi}{4}+\theta\right)$

Sol: $= \dfrac{1+\sin 2\theta}{1-\sin 2\theta} = \dfrac{1+\dfrac{2\tan\theta}{1+\tan^2\theta}}{1-\dfrac{2\tan\theta}{1+\tan^2\theta}} \qquad \because \sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$

$= \dfrac{\dfrac{1+\tan^2\theta+2\tan\theta}{1+\tan^2\theta}}{\dfrac{1+\tan^2\theta-2\tan\theta}{1+\tan^2\theta}} = \dfrac{1+\tan^2\theta+2\tan\theta}{\cancel{1+\tan^2\theta}} \times \dfrac{\cancel{1+\tan^2\theta}}{1+\tan^2\theta-2\tan\theta}$

$= \dfrac{1+\tan^2\theta+2\tan\theta}{1+\tan^2\theta-2\tan\theta} = \dfrac{(1+\tan\theta)^2}{(1-\tan\theta)^2} = \left(\dfrac{1+\tan\theta}{1-\tan\theta}\right)^2 = \left(\dfrac{\tan\frac{\pi}{4}+\tan\theta}{1-\tan\frac{\pi}{4}\tan\theta}\right)^2 = \left(\tan\left(\dfrac{\pi}{4}+\theta\right)\right)^2$

$= \tan^2\left(\dfrac{\pi}{4}+\theta\right) = R.H.S$

xv. $\cos^2\dfrac{\pi}{8}+\cos^2\dfrac{3\pi}{8}+\cos^2\dfrac{5\pi}{8}+\cos^2\dfrac{7\pi}{8}=2$

Sol:

L.H.S $= \cos^2\dfrac{\pi}{8}+\cos^2\dfrac{3\pi}{8}+\cos^2\dfrac{5\pi}{8}+\cos^2\dfrac{7\pi}{8}$

$= \left(\dfrac{1+\cos 2\left(\frac{\pi}{8}\right)}{2}\right)+\left(\dfrac{1+\cos 2\left(\frac{3\pi}{8}\right)}{2}\right)+\left(\dfrac{1+\cos 2\left(\frac{5\pi}{8}\right)}{2}\right)+\left(\dfrac{1+\cos 2\left(\frac{7\pi}{8}\right)}{2}\right)$

$= \dfrac{1+\cos\frac{\pi}{4}}{2}+\dfrac{1+\cos\frac{3\pi}{4}}{2}+\dfrac{1+\cos\frac{5\pi}{4}}{2}+\dfrac{1+\cos\frac{7\pi}{4}}{2} = \dfrac{1+\cos\frac{\pi}{4}+1+\cos\frac{3\pi}{4}+1+\cos\frac{5\pi}{4}+1+\cos\frac{7\pi}{4}}{2}$

$= \dfrac{1}{2}\left(1+\dfrac{1}{\cancel{\sqrt{2}}}+1-\dfrac{1}{\cancel{\sqrt{2}}}+1-\dfrac{1}{\cancel{\sqrt{2}}}+1+\dfrac{1}{\cancel{\sqrt{2}}}\right)$

$= \dfrac{1}{2}(4) = 2 = R.H.S$

$\because \cos\dfrac{\pi}{4} = \cos 45^\circ = \dfrac{1}{\sqrt{2}} \qquad \because \cos\dfrac{3\pi}{4} = \cos 135^\circ = -\dfrac{1}{\sqrt{2}}$

$\because \cos\dfrac{5\pi}{4} = \cos 225^\circ = -\dfrac{1}{\sqrt{2}} \qquad \because \cos\dfrac{7\pi}{4} = \cos 315^\circ = \dfrac{1}{\sqrt{2}}$

Q3. Show that: $2\cos\theta = \sqrt{2+\sqrt{2+2\cos 4\theta}}$

Sol: $2\cos\theta = \sqrt{2+\sqrt{2+2\cos 4\theta}}$

R.H.S $= \sqrt{2+\sqrt{2+2\cos 4\theta}}$

$= \sqrt{2+\sqrt{2(1+\cos 4\theta)}} = \sqrt{2+\sqrt{2(2\cos^2 2\theta)}} = \sqrt{2+\sqrt{4\cos^2 2\theta}} = \sqrt{2+2\cos 2\theta} = \sqrt{2(1+\cos 2\theta)}$

$= \sqrt{2(2\cos^2\theta)} = \sqrt{4\cos^2\theta} = 2\cos\theta = L.H.S$

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Q4. Reduce $\sin^4\theta$ to an expression involving only function of multiples of θ, raised to the first power.

Sol. $\text{Sin}^4\theta = (\text{Sin}^2\theta)^2 = \left(\frac{1-Cos2\theta}{2}\right)^2$ $\quad \boxed{Sin^2\theta = \frac{1-\cos 2\theta}{2}}$ $\quad \boxed{Cos^2 2\theta = \frac{1+\cos 4\theta}{2}}$

$$\frac{1-2Cos2\theta+Cos^2 2\theta}{4} = \frac{1}{4}\left[1-2Cos2\theta+\frac{1+Cos4\theta}{2}\right]$$

$$= \frac{1}{4}\left[\frac{2-4Cos2\theta+1+Cos4\theta}{2}\right] = \frac{1}{8}\,[3 - 4\,\text{Cos}2\theta + \text{Cos}4\theta] = \frac{3-4Cos2\theta+Cos4\theta}{8}$$

Q5. Find the values of $\sin\theta$ and $\cos\theta$ without using table or calculator when θ is.

(i) 18° (ii) 36° (iii) 54° (iv) 72°

Prove that $\cos 36^\circ \cos 72^\circ \cos 108^\circ \cos 144^\circ = \frac{1}{16}$

(i) **When $\theta = 18^\circ$** **Multiply by 5**

Sol: then $5\theta = 90^\circ \Rightarrow 2\theta + 3\theta = 90^\circ \Rightarrow 2\theta = 90^\circ - 3\theta$

$\text{Sin}(2\theta) = \text{Sin}(90^\circ - 3\theta) \Rightarrow 2\,\text{Sin}\theta\,\text{Cos}\theta = \text{Cos}3\theta$

$2\text{Sin}\theta\,\cancel{\text{Cos}\theta} = 4\text{Cos}^3\theta - 3\text{Cos}\theta = \cancel{\text{Cos}\theta}\,(4\text{Cos}^2\theta - 3)$

$2\text{Sin}\theta = 4(1 - \text{Sin}^2\theta) - 3 \Rightarrow 2\text{Sin}\theta = 4 - 4\text{Sin}^2\theta - 3$

$4\text{Sin}^2\theta + 2\text{Sin}\theta - 1 = 0$ $\qquad a = 4,\ b = 2,\ c = -1$

$$\text{Sin}\theta = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-2 \pm \sqrt{(2)^2-4(4)(-1)}}{2(4)}$$

$$\text{Sin}\theta = \frac{-2 \pm \sqrt{4+16}}{8} = \frac{-2 \pm \sqrt{20}}{8} = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{\cancel{2}\,(-1 \pm \sqrt{5})}{\cancel{8}_4}$$

$\text{Sin}\theta = \frac{-1 \pm \sqrt{5}}{4}$ Put $\theta = 18^\circ$ then $\boxed{\text{Sin}\,18^\circ = \frac{-1+\sqrt{5}}{4}}$ because 18° is in I quadrant

$$\text{Cos}^2\theta = 1 - \text{Sin}^2\theta = 1 - \left(\frac{-1+\sqrt{5}}{4}\right)^2 = 1 - \left(\frac{\sqrt{5}-1}{4}\right)^2 = 1 - \left(\frac{5+1-2\sqrt{5}}{16}\right) = \frac{16-6+2\sqrt{5}}{16} = \frac{10+2\sqrt{5}}{16}$$

$$\text{Cos}^2\theta = \frac{10+2\sqrt{5}}{16} \Rightarrow Cos\theta = \pm\frac{\sqrt{10+2\sqrt{5}}}{4} \Rightarrow \boxed{Cos18^\circ = \pm\frac{\sqrt{10+2\sqrt{5}}}{4}}$$

$\text{Cos}18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$ (Because 18° in I quad)

ii. **When $\theta = 36^\circ$**

Sol. then $\text{Cos}2\theta = 2\text{Cos}^2\theta - 1$

Put $\theta = 18^\circ \Rightarrow \text{Cos}2(18^\circ) = 2\text{Cos}^2(18^\circ) - 1$

$$\text{Cos}36^\circ = 2\left(\frac{\sqrt{10+2\sqrt{5}}}{4}\right)^2 - 1 = 2\left(\frac{10+2\sqrt{5}}{16}\right) - 1 = \frac{10+2\sqrt{5}-8}{8} = \frac{2+2\sqrt{5}}{8} = \frac{2(1+\sqrt{5})}{8}$$

$\boxed{\text{Cos}36^\circ = \frac{1+\sqrt{5}}{4}}$

$\text{Sin}^2\theta = 1 - \text{Cos}^2\theta \Rightarrow \text{Sin}^2 36^\circ = 1 - \text{Cos}^2 36^\circ$

$$\text{Sin}^2 36^\circ = 1 - \left(\frac{1+\sqrt{5}}{4}\right)^2 = 1 - \left(\frac{1+5+2\sqrt{5}}{16}\right) = \frac{16-6-2\sqrt{5}}{16} = \frac{10-2\sqrt{5}}{16} \Rightarrow \boxed{Sin36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}}$$

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iii. When $\theta = 54°$

Sol. $\text{Cos}54° = \text{Sin}(90° - 54°) = \text{Sin}36° = \dfrac{\sqrt{10 - 2\sqrt{5}}}{4}$

$\text{Sin}54° = \text{Cos}(90° - 54°) = \text{Cos}36° = \dfrac{1+\sqrt{5}}{4}$

iv. When $\theta = 72°$

Sol. $\text{Sin}72° = \text{Sin}(90° - 18°) = \text{Cos}18° = \dfrac{\sqrt{10+2\sqrt{2}}}{4}$

$\text{Cos}\,72° = \text{Cos}(90° - 18°) = \text{Sin}\,18° = \dfrac{-1+\sqrt{5}}{4}$, ALSO $\text{Cos}36°\,\text{Cos}72°\,\text{Cos}108°\,\text{Cos}144° = \dfrac{1}{16}$

Sol. $\text{L.H.S} = \text{Cos}36°\,\text{Cos}72°\,\text{Cos}108°\,\text{Cos}144°$

$= \text{Cos}36°\,\text{Cos}72°\,\text{Cos}(180° - 72°)\text{Cos}(180° - 36°)$

$= \text{Cos}36°\,\text{Cos}72°\,(-\text{Cos}72°)\,(-\text{Cos}36°) = \text{Cos}^2 36°\,\text{Cos}^2 72°$

$= \left(\dfrac{1+\sqrt{5}}{4}\right)^2 \left(\dfrac{\sqrt{5}-1}{4}\right)^2 = \left(\dfrac{1+5+2\sqrt{5}}{16}\right)\left(\dfrac{5+1-2\sqrt{5}}{16}\right) = \dfrac{(6+2\sqrt{5})(6-2\sqrt{5})}{16\times 16} = \dfrac{(6)^2-(2\sqrt{5})^2}{16\times 16}$

$= \dfrac{36-(4\times 5)}{16\times 16} = \dfrac{36-20}{16\times 16} = \dfrac{16}{16\times 16} = \dfrac{1}{16} = \text{R.H.S}$

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